End semester project review – October 2026

22AIE450 Speech Processing

Speech-to-Action: A Comparative Study of Classical and Deep Learning Approaches for Spoken Intent Classification

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Date of the Presentation: 16-10-2026

Outline

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Introduction and Motivation

- Voice-controlled devices need the intended action, not a transcript
- Conventional spoken language understanding: speech recogniser $\rightarrow$ text classifier
- End-to-end models map audio directly to intent, without transcription
- Published results on Fluent Speech Commands rely on pre-trained encoders
- Contribution of each layer of speech representation is rarely isolated
- This work: one task, four representation tiers, ordered classical $\rightarrow$ modern
- Application context: smart-home commands for lights, heating, music and language

Problem Definition

- **Input:** one spoken command, 16 kHz mono, median duration 2.2 s
- **Output:** intent = (action, object, location), 31 classes
- **Approach:** vary the representation; hold classifier and protocol fixed where possible
- **Tier 1:** hand-crafted features, utterance statistics → SVM / MLP
- **Tier 2:** MFCC sequences → dynamic time warping, Gaussian HMM
- **Tier 3:** MFCC / log-mel sequences → 1D-CNN trained from scratch
- **Tier 4:** frozen wav2vec 2.0 frames → pooled classifier or 1D-CNN
- **Reduced-class tasks:** action (6), object (14), location (4)
- **Metric:** accuracy on speakers unseen in training

Literature Review (1/3): Classical Speech Analysis

Sl.	Title and year	Venue	Methodology	Key findings	Limitations
[1]	"An algorithm for determining the endpoints of isolated utterances" (1975)	Bell Syst. Tech. J.	Energy and ZCR with adaptive thresholds	Reliable endpoints for isolated words	Weak on low-energy fricatives and stops
[2]	"Linear prediction: A tutorial review" (1975)	Proc. IEEE	All-pole model; autocorrelation method, Levinson–Durbin	Unified LP theory; $O(p^2)$ solution	Assumes stationarity; no spectral zeros
[3]	"A pattern recognition approach to voiced-unvoiced-silence classification …" (1976)	IEEE TASSP	Statistical classifier on energy, ZCR, ACF, LPC	Joint features beat single thresholds	Small, clean recordings
[4]	"On the use of autocorrelation analysis for pitch detection" (1977)	IEEE TASSP	Short-time ACF with spectral flattening	Restricted lag range curbs octave errors	Errors persist at weak periodicity
[5]	"A unified approach to short-time Fourier analysis and synthesis" (1977)	Proc. IEEE	STFT analysis–synthesis theory	Window length sets time–frequency trade-off	Theoretical; no recognition results
[6]	"YIN, a fundamental frequency estimator for speech and music" (2002)	J. Acoust. Soc. Am.	Cumulative mean-normalised difference	Fewer gross errors than ACF	Needs voicing threshold; no tracking
[7]	"Comparison of parametric representations for monosyllabic word recognition …" (1980)	IEEE TASSP	MFCC vs. LPC-derived parameters	MFCC best: 96.5% / 95.0% (two speakers)	Two speakers; coefficients not interpretable

Literature Review (2/3): Sequence and Neural Models

Sl.	Title and year	Venue	Methodology	Key findings	Limitations
[8]	"Dynamic programming algorithm optimization for spoken word recognition" (1978)	IEEE TASSP	DTW with slope constraints	Optimal alignment of unequal durations	Cost grows with number of templates
[9]	"Speaker-independent isolated word recognition using dynamic features . . . " (1986)	IEEE TASSP	Delta (time-derivative) cepstra	Dynamic features reduce error markedly	Isolated words, small vocabulary
[10]	"A tutorial on hidden Markov models and selected applications . . . " (1989)	Proc. IEEE	Forward–backward, Viterbi, Baum–Welch	Standard statistical sequence model	Topology and mixtures need tuning
[11]	"Understanding how deep belief networks perform acoustic modelling" (2012)	IEEE ICASSP	DBN on filter bank vs. MFCC	Filter-bank input outperforms MFCC	Phone recognition, not intent
[12]	"Convolutional neural networks for speech recognition" (2014)	IEEE/ACM TASLP	CNN over filter-bank features	Lower error than fully connected DNNs	Needs frequency-local input, not MFCC
[13]	"Convolutional neural networks for small-footprint keyword spotting" (2015)	Interspeech	Small CNN on log-mel features	27–44% relative false-reject reduction vs. DNN	Keywords, not multi-slot intents
[14]	"SpecAugment: A simple data augmentation method for ASR" (2019)	Interspeech	Time warping, time and frequency masking	6.8% WER on LibriSpeech test-other, no LM	Tuned for large ASR models

Literature Review (3/3): End-to-End SLU and SSL

Sl.	Title and year	Venue	Methodology	Key findings	Limitations
[15]	"Towards end-to-end spoken language understanding" (2018)	IEEE ICASSP	Audio-to-intent model without ASR	Direct intent prediction is feasible	Needs large labelled data; opaque
[16]	"Speech model pre-training for end-to-end spoken language understanding" (2019)	Interspeech	ASR-pre-trained layers; releases FSC	98.8% pre-trained; 96.6% from scratch	Single split with 10 test speakers
[17]	"wav2vec 2.0: A framework for self-supervised learning of speech representations" (2020)	NeurIPS	Masked contrastive learning on raw audio	Strong ASR with little labelled data	Large model; fine-tuning is costly
[18]	"End-to-end neural transformer based spoken language understanding" (2020)	Interspeech	Transformer without intermediate tokens	99.6% intent accuracy on FSC	No analysis of features or errors
[19]	"HuBERT: Self-supervised speech representation learning by masked prediction …" (2021)	IEEE/ACM TASLP	Masked prediction of k-means units	Matches or exceeds wav2vec 2.0 on ASR	Iterative clustering adds cost
[20]	"SUPERB: Speech processing universal performance benchmark" (2021)	Interspeech	Frozen encoders with light task heads	FSC: FBANK 9.1%, wav2vec 2.0 92.4%, HuBERT 98.3%	Fixed heads; official split only
[21]	"Layer-wise analysis of a self-supervised speech representation model" (2021)	IEEE ASRU	CCA and probes per wav2vec 2.0 layer	Phone and word content peaks in middle layers	Analysis study; ASR-oriented

Gaps Identified

1. FSC literature is dominated by pre-trained end-to-end models [16], [18], [20]
2. Prosodic vs. cepstral gap on an intent task not measured with a fixed classifier [7]
3. DTW/HMM and neural sequence models rarely compared on the same inputs [8], [10], [12]
4. Effect of MFCC's DCT step on a CNN untested for intent classification [11]
5. Sensitivity of FSC results to its ten-speaker test split is not reported [16], [20]
6. Residual errors of strong SLU models are seldom analysed beyond accuracy

Objectives

- Implement and verify a classical short-time analysis front end
- Measure intent accuracy of prosodic vs. cepstral utterance statistics
- Benchmark DTW and Gaussian HMM classifiers on MFCC sequences
- Train 1D-CNNs from scratch on MFCC and log-mel sequences
- Evaluate frozen wav2vec 2.0 with pooled and sequence classifiers
- Repeat on reduced-class slot tasks: 4, 6 and 14 classes
- Compare all models under speaker-grouped cross-validation
- Analyse residual errors by slot, intent pair and speaker

Dataset Description

- Fluent Speech Commands (FSC) [16], academic licence
- 30,043 utterances, 19.0 h, 97 speakers, 16 kHz mono
- Slots: action (6), object (14), location (4) $\rightarrow$ 31 intents
- 248 distinct phrasings, e.g. “Turn the lamp off”
- Split: 23,132 / 3,118 / 3,793 utterances
- Speakers: 77 / 10 / 10, no overlap (verified)
- Imbalance: 2,296 vs. 342 training utterances per intent
- Duration: mean 2.31 s, 95th percentile 3.41 s

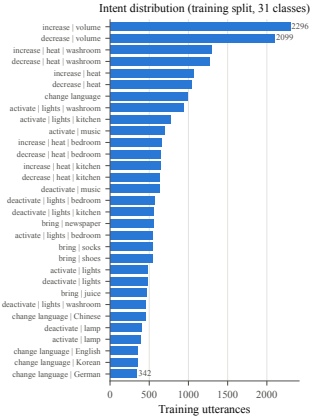


Figure 1: Training utterances per intent.

Methodology: Overall Flow

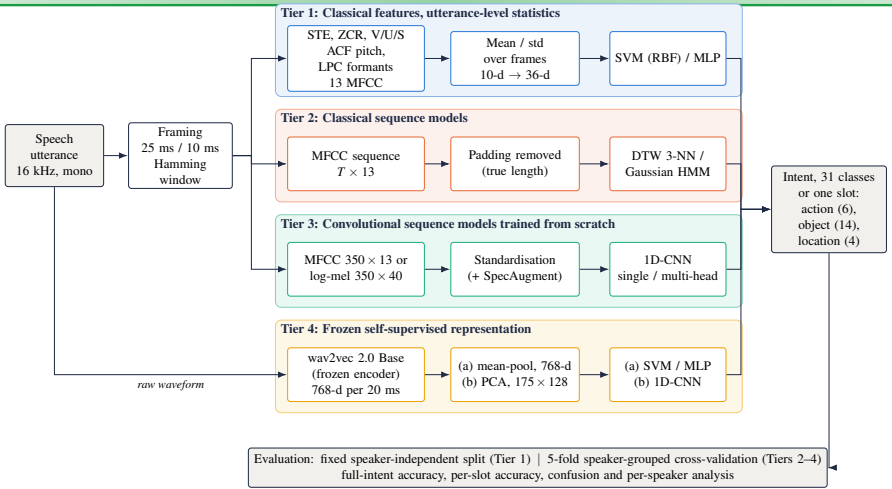


Figure 2: Four representation tiers, ordered from classical to modern.

Methodology: Classical Front End

- Framing: 25 ms frames (400 samples), 10 ms hop, Hamming window
- Short-time energy $E_m = \sum_n x_m^2[n]$; ZCR = fraction of sign changes per frame
- Speech if energy $>$ utterance noise floor + 10 dB; unvoiced if ZCR $>$ 70th percentile
- Pitch: autocorrelation peak in 70–400 Hz; confidence $r[k^*]/r[0] \geq 0.3$
- Linear prediction: order 16, autocorrelation method, Levinson–Durbin recursion
- MFCC: mel filter bank $\rightarrow$ log $\rightarrow$ DCT $\rightarrow$ 13 coefficients
- Log-mel: same pipeline without the DCT, 40 bands
- Utterance vector: mean / std statistics, 10-d (prosodic) and 36-d (+ MFCC)
- Full-corpus pitch with YIN [6]; same periodicity principle, faster

Methodology: Models and Evaluation Protocol

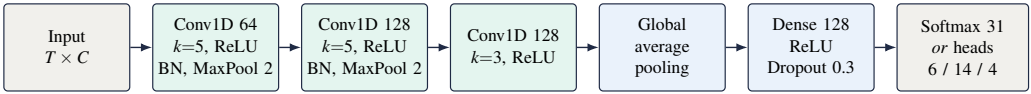


Figure 3: 1D-CNN shared by Tiers 3 and 4 (115,871 parameters on MFCC input).

- Tier 1: SVM (RBF) and MLP, grid-searched on the official validation subset
- Tier 2: DTW 3-NN with 5 templates per intent; 3-state diagonal Gaussian HMM per intent
- Tier 3: Adam, batch 128, early stopping; multi-head variant adds SpecAugment [14]
- Tier 4: frozen wav2vec 2.0 Base [17]; mean-pooled 768-d or PCA-128 sequence
- Evaluation: same 5 speaker folds for every model and task

Results: Time-Domain Analysis of “Bring socks”

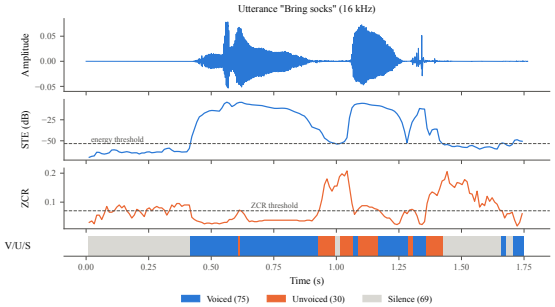


Figure 4: Waveform, STE, ZCR and V/U/S labels.

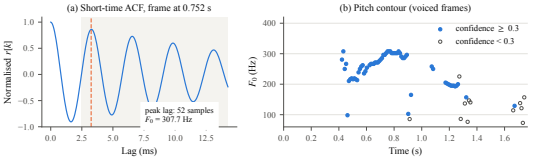


Figure 5: ACF of a voiced frame and F_0 contour.

- Frames: 75 voiced, 30 unvoiced, 69 silent
- /s/: low energy with ZCR of 0.15–0.21
- /k/ burst shorter than one 25 ms frame
- F_0 210 $\rightarrow$ 310 Hz on “bring”
- 63 of 75 reliable; octave error at 0.463 s

Results: Spectral Analysis and Linear Prediction

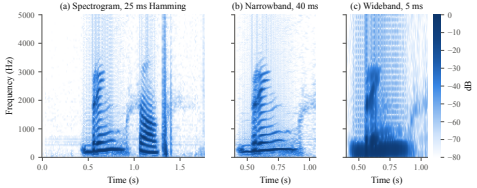


Figure 6: Spectrogram; narrowband vs. wideband.

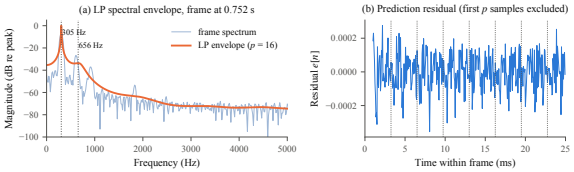


Figure 7: LP envelope ($p = 16$) and residual.

- Harmonic spacing agrees with ACF pitch
- Fricatives: diffuse energy, no harmonics
- /k/ release: brief broadband stripe
- 40 ms resolves harmonics; 5 ms, pulses
- LP resonances at 305 Hz and 656 Hz
- Burg-method envelope: same peaks
- Residual periodicity falls as p rises

Results: Tier 1, Utterance-Level Statistics

Table 1: 31-class accuracy (%): official split and 5-fold speaker CV.

Features	Classifier	Valid.	Test	CV
–	Majority	10.2	10.4	10.0 ± 0.2
Prosodic	SVM	16.4	16.5	17.8 ± 1.0
Prosodic	MLP	15.2	14.3	16.6 ± 1.5
+ MFCC	SVM	22.3	**25.8**	$\mathbf{24.0 \pm 1.0}$
+ MFCC	MLP	22.4	21.0	21.4 ± 1.0

- Prosody alone: about 7 points above the majority class
- Tuning plateaus near 15–16%: features limit accuracy
- + MFCC: +6.2 points in every CV fold ($p < 0.001$)
- Volume intents: 20% of outputs vs. 10.4% of labels
- CV confirms the fixed-split conclusions

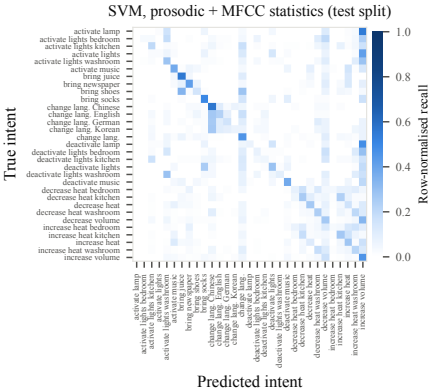


Figure 8: SVM on prosodic + MFCC statistics.

Results: Tier 2, Classical Sequence Models

Table 2: Accuracy (%) of the classical sequence classifiers.

Model	Accuracy	Evaluation
Majority class	10.4	official test split
DTW, 3-nearest neighbours	11.6	155 queries, one fold
Gaussian HMM, padded input	9.8 ± 0.1	5-fold speaker-grouped CV
Gaussian HMM, padding removed	6.0 ± 0.7	5-fold speaker-grouped CV

- Both baselines lie near or below the majority class
- Padded HMM exploited padding length, an indirect duration cue
- After padding removal, spectral discrimination alone gives 6.0%
- HMM kept minimal: 3 ergodic states, one Gaussian, no delta features
- DTW used only 5 templates per intent against 248 phrasings
- Result reflects the minimal configuration, not the classical method itself

Results: Tier 3, CNNs Trained from Scratch

Table 3: 5-fold speaker-grouped CV, 31-class accuracy (%).

Input / model	Dim.	Mean ± s.d.
1. Input representation (single head)		
MFCC	13	72.5 ± 5.0
Log-mel	40	**79.4 ± 6.8**
DCT of log-mel, first 13	13	73.1 ± 5.8
DCT of log-mel, all 40	40	63.5 ± 5.0
2. Output structure (MFCC input)		
Multi-head + SpecAugment	13	77.3 ± 6.7

- Same MFCC frames, order kept: 24.0 → 72.5%
- Log-mel beats MFCC in every fold (+6.9, $p = 0.005$)
- Full DCT: −15.9 points (weak coefficients inflated)

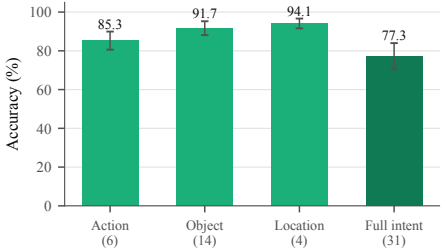


Figure 9: Multi-head CNN accuracy per slot.

- Multi-head: +4.8 over MFCC, every fold
- Action is the weakest slot (85.3%)
- vs. log-mel: n.s. ($p = 0.16$)

Results: Tier 4, Frozen wav2vec 2.0

Table 4: This work (5-fold speaker-grouped CV).

Downstream classifier	Acc. (%)
Mean-pooled + SVM	59.9 ± 4.4
Mean-pooled + MLP	69.9 ± 4.7
Frame sequence + 1D-CNN	$\mathbf{93.1 \pm 3.1}$

Table 5: Published FSC results (official test split).

Model	Acc. (%)
wav2vec 2.0 Base, frozen [20]	92.35
HuBERT Base, frozen [20]	98.34
ASR pre-trained, frozen [16]	98.8
End-to-end transformer [18]	99.6

- Pooling discards order: pooled MLP $\approx$ MFCC CNN
- Frame sequence + CNN: best model in every fold
- +13.7 points over the log-mel CNN ($p = 0.001$)
- Lowest fold-to-fold spread of all models
- Close to SUPERB frozen wav2vec 2.0 Base (92.35%)
- Gap to 98–99.6%: last layer only [21], PCA, frozen

Results: Overall Comparison

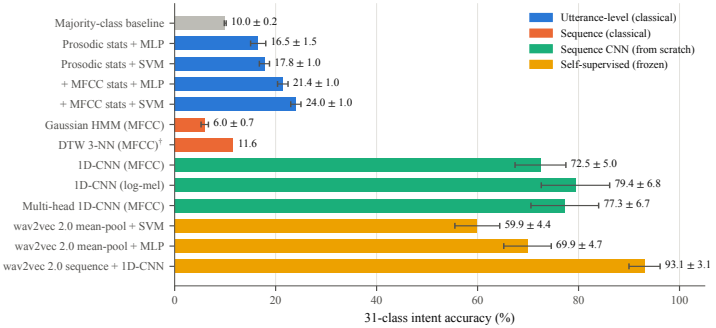


Figure 10: All configurations under 5-fold speaker CV, classical to modern (†155-query subset).

- Cepstral content: 17.8 → 24.0%
- Temporal order: 24.0 → 72.5–79.4%
- Self-supervised frames: 79.4 → 93.1%
- All tiers on the same speaker folds

Results: Reduced-Class Slot Tasks

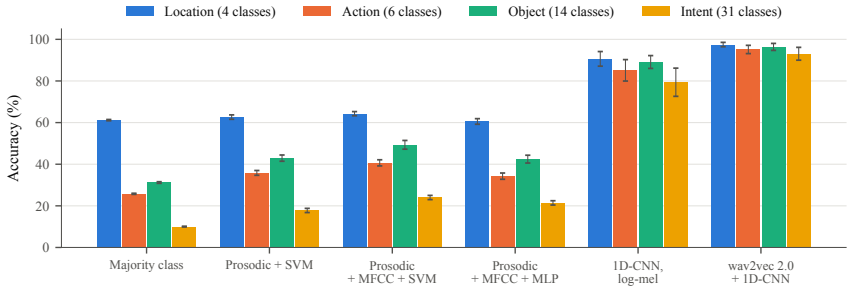


Figure 11: Accuracy per task under 5-fold speaker CV (classes: location 4, action 6, object 14, intent 31).

- Fewer classes raise accuracy for every model
- Majority baseline also rises: 10.0 → 25.8–61.2%
- SVM removes only 8–26% of the majority error
- Location: MLP below majority (60.5 vs. 61.2%)
- CNNs remove 76–95% of the majority error
- Limit is the representation, not the class count

Results: Error and Speaker Variability Analysis

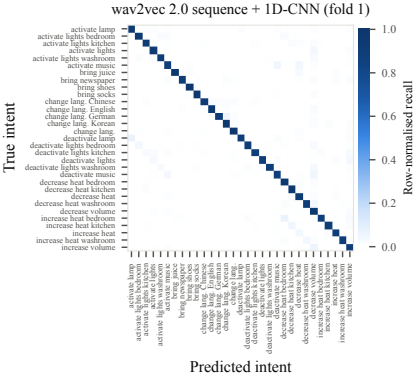


Figure 12: wav2vec 2.0 + CNN, fold 1 (91.7%).

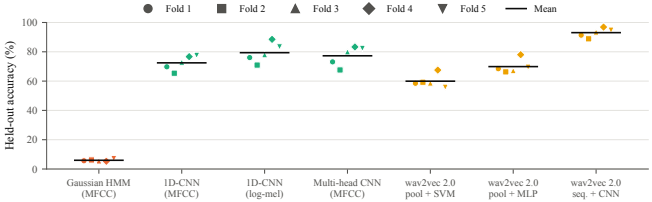


Figure 13: Held-out accuracy per fold.

- Slot accuracy: action 93.9, object 95.2, location 97.2%
- Top confusion: increase → decrease volume (40 errors)
- Polarity often rests on one particle: *up/down*, *on/off*
- Speakers: 36.4–100% accuracy; 14 of 19 above 89%

Conclusion and Future Work

Conclusion

- Pitch, spectral and LP analyses agree with each other
- Prosodic statistics alone give only 17.8%
- MFCC statistics recover part of the gap (24.0%)
- Keeping temporal order gives the largest gain
- DCT + standardisation hurts the CNN; log-mel is better
- Frozen wav2vec 2.0 + CNN is best: $93.1 \pm 3.1\%$
- Slot tasks: classical stays near majority; CNNs 85–97%
- Residual errors: opposite-polarity commands
- FSC needs speaker-grouped CV, not one split

Future Work

- Left-to-right GMM-HMM with deltas [9]
- Learned wav2vec 2.0 layer weighting [20]
- HuBERT and encoder fine-tuning [19]
- Robustness to noise and unseen phrasings

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